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# Cyclic codes over $M_4(\mathbb{F}_2 + u\mathbb{F}_2)$

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**Abstract**
Let $p$ be a prime and $\mathbb{F}_q$ be a finite field for $q = p^m$. In this paper, we consider the ring $R = M_4(\mathbb{F}_2 + u\mathbb{F}_2)$ of $4 \times 4$ matrices over the finite ring $\mathbb{F}_2 + u\mathbb{F}_2$ with $u^2 = 0$. Then $R$ is a noncommutative non-chain ring of cardinality $4^{16}$ and isomorphic to the ring $\mathbb{F}_{16} + v\mathbb{F}_{16} + v^2\mathbb{F}_{16} + v^3\mathbb{F}_{16} + u\mathbb{F}_{16} + uv\mathbb{F}_{16} + uv^2\mathbb{F}_{16} + uv^3\mathbb{F}_{16}$, where $v^4 = 0$, $uv = vu$, $uv^2 = v^2u$ and $uv^3 = v^3u$. Here, first we establish the structure of cyclic codes and their generators over $R$ and later the dual (Euclidean and Hermitian both) of these cyclic codes are discussed. Further, with the help of the Gray map, we show that the image of a cyclic code is an $\mathbb{F}_{16}$-linear code. Finally, we provide some non-trivial examples of linear codes with good parameters to support our derived results.

**Keywords** Modules · Matrix ring · Cyclic codes · Gray map

**Mathematics subject classification (2010)** 12E20 · 94B05 · 94B15

## 1 Introduction

The study of linear codes over finite rings has received much attention in the last decades of the twentieth century. A linear code whose codewords are invariant under a circular shift is known as a *cyclic code*. A cyclic code of length $n$ over a ring $R$ is an ideal of the quotient ring $R[x]/\langle x^n - 1\rangle$, and each codeword is identified by a polynomial of degree at most $n - 1$. Due to such algebraic structure, the class of cyclic codes is a better choice among researchers for encoding and decoding purposes than other linear codes [1, 5]. In 1994, Hammons et al. [9] gave a new direction to linear codes over $\mathbb{Z}_4$ and obtained some good non-linear binary codes as the Gray images of linear codes. Afterwards, the study of linear codes over finite rings has gained significant attention from the coding analysts.

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Besides several remarkable studies over finite commutative rings, recently codes over noncommutative rings have been attracting many researchers, we refer [2, 8, 11–13, 15]. In fact, in 1997, the first concrete set of alphabets $A = M_2(\mathbb{F}_2)$ appeared in algebraic constructions of modular lattices [3]. Later, in 2012, Oggier et al. [12] developed the theory for noncommutative ring $M_2(\mathbb{F}_2)$ in the context of space-time codes. In 2013, Alahmadi et al. [2] studied cyclic codes over the same ring $M_2(\mathbb{F}_2)$. In 2014, Falcunit and Sison [7] discussed the structure of cyclic codes over $M_2(\mathbb{F}_p)$ and obtained the isometric images of these codes over the ring $\mathbb{F}_{p^2} + u\mathbb{F}_{p^2}$. Moreover, in 2018, Luo and Udaya [11] presented the structure of cyclic codes over the matrix ring $M_2(\mathbb{F}_2 + u\mathbb{F}_2)$ and obtained some optimal codes of even length over $\mathbb{F}_4$. Again, in 2018, Bhowmick et al. [4] discussed cyclic codes and their dual over a matrix ring $M_2(\mathbb{Z}_4)$. Further, in 2019, Pal et al. [13] studied cyclic codes over the matrix ring $M_4(\mathbb{F}_2)$. Recently, Islam et al. [10] extended the work of [11] to the ring $M_2(\mathbb{F}_p + u\mathbb{F}_p)$, where $p \equiv 2$ or $3 \pmod 5$ and studied cyclic codes of length $n$.

Throughout this paper, $R = M_4(\mathbb{F}_2 + u\mathbb{F}_2)$, a noncommutative ring of $4 \times 4$ matrices over $\mathbb{F}_2 + u\mathbb{F}_2$ with $u^2 = 0$. It is easy to check that $R = M_4(\mathbb{F}_2) + uM_4(\mathbb{F}_2) \cong M_4(\mathbb{F}_2)[u]/\langle u^2 \rangle$. From [13], if we consider

$$e = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}$$

and

$$\omega = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \end{pmatrix},$$

then $M_4(\mathbb{F}_2)$ is isomorphic to the ring $\mathbb{F}_{16} + e\mathbb{F}_{16} + e^2\mathbb{F}_{16} + e^3\mathbb{F}_{16}$, where $e$ satisfies the relation $(1 + e)^4 = 0$ and $\mathbb{F}_{16}$ is isomorphic to $\mathbb{F}_2[\omega]$. Moreover, $\mathbb{F}_{16} + e\mathbb{F}_{16} + e^2\mathbb{F}_{16} + e^3\mathbb{F}_{16}$ can also be represented as $\mathbb{F}_{16} + v\mathbb{F}_{16} + v^2\mathbb{F}_{16} + v^3\mathbb{F}_{16}$ with $v^4 = (1 + e)^4 = 0$, i.e., $v$ is a nilpotent element of nilpotency 4. Hence, $R = \mathbb{F}_{16} + v\mathbb{F}_{16} + v^2\mathbb{F}_{16} + v^3\mathbb{F}_{16} + u\mathbb{F}_{16} + uv\mathbb{F}_{16} + uv^2\mathbb{F}_{16} + uv^3\mathbb{F}_{16}$, where $v^4 = 0, u^2 = 0, uv = vu, uv^2 = v^2u$ and $uv^3 = v^3u$.

In this paper, we focus on cyclic codes of odd length over $R = M_4(\mathbb{F}_2 + u\mathbb{F}_2)$ where $u^2 = 0$. This also enables the construction of cyclic codes of even length over $\mathbb{F}_{16}$ with the help of the Gray map. Further, we define the Bachoc map from $M_4(\mathbb{F}_2 + u\mathbb{F}_2)$ to $M_4^2(\mathbb{F}_2)$ which preserves the Bachoc weight. We observed that Pal et al. [13] have discussed results of cyclic codes only for odd lengths. However, we extend it and also obtain the images of cyclic codes of odd length over $R$ under the Bachoc map as binary quasi-cyclic codes of even length over $M_4(\mathbb{F}_2)$. Undoubtedly, Pal et al. [13] have given a purposeful research venture to study the dual of cyclic codes over $M_4(\mathbb{F}_2)$ and we extend this study to the Euclidean dual as well as the Hermitian dual of cyclic codes over $R$. A major motivation to study the Hermitian dual codes is that these codes have been efficiently used in the construction of modular lattices [3].

The presentation of the paper is as follows: Section 2 discusses the ring structure of $R = M_4(\mathbb{F}_2 + u\mathbb{F}_2)$ and the characterization of cyclic codes over it. Sections 3 and 4 contain the Euclidean dual and Hermitian dual of cyclic codes over $R$, respectively. In Section 5, we define the Gray map and the Bachoc map to determine a correspondence between

$M_4(\mathbb{F}_2 + u\mathbb{F}_2)$ and $\mathbb{F}_{16}^8$; $M_4(\mathbb{F}_2 + u\mathbb{F}_2)$ and $M_4^2(\mathbb{F}_2)$, respectively. Section 6 presents some examples of non-trivial codes with good parameters in support of our obtained results. Section 7 concludes the paper.

## 2 Cyclic codes over $M_4(\mathbb{F}_2 + u\mathbb{F}_2)$

A linear code $C$ of length $n$ over $R$ is an $R$-submodule of $R^n$. It is said to be cyclic if for each codeword $c = (c_0, c_1, \ldots, c_{n-1}) \in C$, its cyclic shift $(c_{n-1}, c_0, \ldots, c_{n-2}) \in C$. Also, a cyclic code of length $n$ over $R$ can be viewed as an ideal of $R_n = R[x]/\langle x^n - 1\rangle$, and each codeword $(c_0, c_1, \ldots, c_{n-1})$ is identified by a polynomial $c_0 + c_1 x + \cdots + c_{n-1}x^{n-1}$.

Let $R[x]$ be the ring of polynomials over $R$. For any $a \in R$, let $\tilde{a}$ be the polynomial reduction of $a$ modulo $u$ and $v$. We consider a polynomial reduction mapping $\rho : R[x] \to \mathbb{F}_{16}[x]$ defined by

$$\sum_{i=0}^{n-1} a_i x^i \to \sum_{i=0}^{n-1} \tilde{a}_i x^i.$$

A polynomial $f$ is called basic irreducible over $R$ if $\rho(f)$ is irreducible over $\mathbb{F}_{16}$. If $n$ is odd and $x^n - 1 = \prod_{r=1}^{m} f_r$, where $f_r$'s are irreducible polynomials over $\mathbb{F}_{16}$, then the polynomial $x^n - 1$ can be uniquely factorized into pairwise coprime polynomials over $\mathbb{F}_{16}$.

**Theorem 1** *Let $n$ be an odd positive integer and $x^n - 1 = \prod_{r=1}^{m} f_r$, where $f_r$'s are basic irreducible polynomials over $R$. Then*

$$R_n = R[x]/\langle x^n - 1\rangle = \bigoplus_{r=1}^{m} R[x]/\langle f_r\rangle.$$

***Proof*** Follows from [12, 14] and application of the Chinese Remainder Theorem to the right $R$-module $R[x]/\langle x^n - 1\rangle$.

The above theorem is an analogue of the Chinese Remainder Theorem for noncommutative setup. Since $\langle f_r\rangle$ is not a two sided ideal, $R[x]/\langle f_r\rangle$ is treated as a right $R$-module. Now, we find all the possible forms of the submodules of $R[x]/\langle f_r\rangle$.

**Theorem 2** *If $f$ is a basic irreducible polynomial over $R$, then the right $R$-submodules of $R_f = R[x]/\langle f\rangle$ are $\langle 0\rangle$, $\langle 1\rangle$, $\langle v\rangle$, $\langle v^2\rangle$, $\langle v^3\rangle$, $\langle uv\rangle$, $\langle uv^2\rangle$, $\langle uv^3\rangle$, $\langle u\rangle$, $\langle uv^2\rangle + \langle v^3\rangle$, $\langle uv + v^2h_\alpha + v^3h_\beta + uv^2h_\delta\rangle$, $\langle u + vh_\alpha + v^2h_\beta + v^3h_\gamma + uvh_\delta + uv^2h_{\delta'}\rangle$ and $\langle uv^2 + v^3h_\alpha\rangle$, where $h_\alpha, h_\beta, h_\gamma, h_\delta, h_{\delta'}$ are units in $\mathbb{F}_{16}[x]$.*

***Proof*** Let $I$ be a non-zero right $R$-submodule of $R_f$ and $g + \langle f\rangle$, a non-zero element of $I$. Then $g \notin \langle f\rangle$ and there exist $g_1, g_2, g_3, g_4, g_5, g_6, g_7, g_8 \in \mathbb{F}_{16}[x]$ such that

$$g = g_1 + vg_2 + v^2g_3 + v^3g_4 + ug_5 + uvg_6 + uv^2g_7 + uv^3g_8. \tag{1}$$

There are two cases:

**Case I** If $\gcd(g_1, f) = 1$, then there exists $g_1' \in \mathbb{F}_{16}[x]$ such that $g_1g_1' + \langle f\rangle = 1 + \langle f\rangle$. This implies that $I = R_f$.

**Case II** If $\gcd(g_1, f) = f$, then

$$g + \langle f \rangle = vg_2 + v^2g_3 + v^3g_4 + ug_5 + uvg_6 + uv^2g_7 + uv^3g_8 + \langle f \rangle. \tag{2}$$

If $\gcd(g_5,f) = f$, then

$$g + \langle f \rangle = vg_2 + v^2g_3 + v^3g_4 + uvg_6 + uv^2g_7 + uv^3g_8 + \langle f \rangle.$$

This implies that $I = \langle v \rangle$ or $\langle v^2 \rangle$ or $\langle v^3 \rangle$ or $\langle uv \rangle$ or $\langle uv^2 \rangle$ or $\langle uv^3 \rangle$.

If $\gcd(g_5,f) = 1$, then there exists $g_5^{-1} \in \mathbb{F}_{16}[x]$ such that $g_5g_5^{-1} = 1$. Now, we have $uv^3 = gv^3g_5^{-1} \in I$. It follows that $ug_5 + vg_2 + v^2g_3 + v^3g_4 + uvg_6 + uv^2g_7 = g - uv^3g_8 \in I$.

If $\gcd(g_2,f) = \gcd(g_3,f) = \gcd(g_4,f) = \gcd(g_6,f) = \gcd(g_7,f) = f$, then $u \in I$ and hence $I = \langle u \rangle$. Further, if $\gcd(g_2,f) = \gcd(g_3,f) = \gcd(g_4,f) = \gcd(g_6,f) = \gcd(g_7,f) = 1$, then $\langle u + vh_\alpha + v^2h_\beta + v^3h_\gamma + uvh_\delta + uv^2h_{\delta'} \rangle \in I$ where $h_\alpha = g_2g_5^{-1}, h_\beta = g_3g_5^{-1}, h_\gamma = g_4g_5^{-1}, h_\delta = g_6g_5^{-1}, h_{\delta'} = g_7g_5^{-1}$ are units in $\mathbb{F}_{16}[x]$. Again, there are two cases:

1. $\langle u + vh_\alpha + v^2h_\beta + v^3h_\gamma + uvh_\delta + uv^2h_{\delta'} \rangle = I$.
2. $\langle u + vh_\alpha + v^2h_\beta + v^3h_\gamma + uvh_\delta + uv^2h_{\delta'} \rangle \subsetneq I$.

From (2), we have

$$gv = v^2g_2 + v^3g_3 + uvg_5 + uv^2g_6 + uv^3g_7.$$

This implies that $gvg_5^{-1} - uv^3g_7g_5^{-1} = v^2g_2g_5^{-1} + v^3g_3g_5^{-1} + uv + uv^2g_6g_5^{-1} \in I$. If $\gcd(g_2,f) = \gcd(g_3,f) = \gcd(g_6,f) = 1$, then $uv + v^2g_2g_5^{-1} + v^3g_3g_5^{-1} + uv^2g_6g_5^{-1} \in I$, where $h_\alpha = g_2g_5^{-1}, h_\beta = g_3g_5^{-1}, h_\delta = g_6g_5^{-1}$ are units in $\mathbb{F}_{16}[x]$. Therefore, we have the following cases:

(i) $\langle uv + v^2h_\alpha + v^3h_\beta + uv^2h_\delta \rangle = I$.
(ii) $\langle uv + v^2h_\alpha + v^3h_\beta + uv^2h_\delta \rangle \subsetneq I$.

Again, from (2), we have

$$gv^2 = v^3g_2 + uv^2g_5 + uv^3g_6.$$

This implies that $v^3g_2 + uv^2g_5 = gv^2 - uv^3g_6 \in I$. If $\gcd(g_2,f) = 1$, then $uv^2 + v^3h_\alpha \in I$, where $h_\alpha = g_2g_5^{-1}$ is a unit in $\mathbb{F}_{16}[x]$. Now, further we have two cases:

(iii) $\langle uv^2 + v^3h_\alpha \rangle = I$.
(iv) $\langle uv^2 + v^3h_\alpha \rangle \subsetneq I$.

Since there exists $uv^2 + v^3h_{\alpha'} \in I$ where $h_{\alpha'} \in \mathbb{F}_{16}[x]$, which does not belong to $\langle uv^2 + v^3h_\alpha \rangle$. This implies that $v^3(h_\alpha - h_{\alpha'}) \in I$. But, $(h_\alpha - h_{\alpha'}) \notin \langle f \rangle$, therefore, $v^3 \in I$. Hence, $I = \langle uv^2 \rangle + \langle v^3 \rangle$.

Let $f_i$ be a factor of $x^n - 1$ and consider $\hat{f}_i = \frac{x^n-1}{f_i}$. Then we have the following result for $R$-modules in $\frac{R[x]}{\langle x^n-1 \rangle}$.

**Corollary 1** *Let $x^n-1=f_1f_2\cdots f_m$ where $f_i(1\le i\le m)$ are irreducible pairwise coprime monic polynomials in $\mathbb{F}_{16}[x]$. Then any right R-module in $\frac{R[x]}{\langle x^n-1\rangle}$ is the sum of the R-submodules $\langle \hat{f}_i+\langle x^n-1\rangle\rangle$, $\langle v\hat{f}_i+\langle x^n-1\rangle\rangle$, $\langle v^2\hat{f}_i+\langle x^n-1\rangle\rangle$, $\langle v^3\hat{f}_i+\langle x^n-1\rangle\rangle$, $\langle uv\hat{f}_i+\langle x^n-1\rangle\rangle$, $\langle uv^2\hat{v}_i+\langle x^n-1\rangle\rangle$, $\langle uv^3\hat{f}_i+\langle x^n-1\rangle\rangle$, $\langle u\hat{f}_i+\langle x^n-1\rangle\rangle$, $\langle(uv+v^2h_\alpha+v^3h_\beta+uv^2h_\delta)\hat{f}_i+\langle x^n-1\rangle\rangle$, $\langle(u+vh_\alpha+v^2h_\beta+v^3h_\gamma+uvh_\delta+uv^2h_{\delta'})\hat{f}_i+\langle x^n-1\rangle\rangle$ and $\langle(uv^2+v^3h_\alpha)\hat{f}_i+\langle x^n-1\rangle\rangle$ for $1\le i\le m$.*

***Proof*** Follows from Theorems 1 and 2.

Throughout the paper, we use $c_0+c_1x+\cdots+c_{n-1}x^{n-1}$ for the corresponding coset $c_0+c_1x+\cdots+c_{n-1}x^{n-1}+\langle x^n-1\rangle$ in the ring $\frac{R[x]}{\langle x^n-1\rangle}$. Now, we provide the main result of this section.

**Theorem 3** *Let $C$ be a cyclic code of odd length $n$ over $R$. Then there exist units $h_\alpha,h_\beta,h_\gamma,h_\delta,h_{\delta'}$ in $\mathbb{F}_{16}[x]$ and a family of irreducible pairwise coprime monic polynomials $G_i$, for $0\le i\le 12$, in $\mathbb{F}_{16}[x]$ such that $\prod_{i=0}^{12}G_i=x^n-1$ and*

$$\begin{aligned}C=&\langle\hat{G}_1\rangle\oplus\langle v\hat{G}_2\rangle\oplus\langle v^2\hat{G}_3\rangle\oplus\langle v^3\hat{G}_4\rangle\oplus\langle uv\hat{G}_5\rangle\oplus\langle uv^2\hat{G}_6\rangle\oplus\langle uv^3\hat{G}_7\rangle\oplus\langle u\hat{G}_8\rangle\oplus(\langle uv^2\hat{G}_9\rangle\\&+\langle v^3\hat{G}_9\rangle)\oplus\langle(uv+v^2h_\alpha+v^3h_\beta+uv^2h_\delta)\hat{G_{10}}\rangle\oplus\langle(u+vh_\alpha+v^2h_\beta+v^3h_\gamma+uvh_\delta\\&+uv^2h_{\delta'})\hat{G_{11}}\rangle\oplus\langle(uv^2+v^3h_\alpha)\hat{G_{12}}\rangle.\end{aligned}$$

*Moreover, $|C|=16^\xi$, where $\xi=8\deg(G_1)+6\deg(G_2)+4\deg(G_3)+2\deg(G_4)+3\deg(G_5)+2\deg(G_6)+\deg(G_7)+4\deg(G_8)+3\deg(G_9)+3\deg(G_{10})+5\deg(G_{11})+2\deg(G_{12})$.*

***Proof*** Let $C$ be a cyclic code of odd length $n$ over $R$ and $x^n-1=f_1f_2\cdots f_m$ where $f_i(1\le i\le m)$ are pairwise coprime monic irreducible polynomials in $\mathbb{F}_{16}[x]$. By Corollary 1, $C$ can be written as the direct sum of the right $R$-submodules $\langle\hat{f}_i+\langle x^n-1\rangle\rangle$, $\langle v\hat{f}_i+\langle x^n-1\rangle\rangle$, $\langle v^2\hat{f}_i+\langle x^n-1\rangle\rangle$, $\langle v^3\hat{f}_i+\langle x^n-1\rangle\rangle$, $\langle uv\hat{f}_i+\langle x^n-1\rangle\rangle$, $\langle uv^2\hat{f}_i+\langle x^n-1\rangle\rangle$, $\langle uv^3\hat{f}_i+\langle x^n-1\rangle\rangle$, $\langle u\hat{f}_i+\langle x^n-1\rangle\rangle$, $\langle(uv+v^2h_\alpha+v^3h_\beta+uv^2h_\delta)\hat{f}_i+\langle x^n-1\rangle\rangle$, $\langle(u+vh_\alpha+v^2h_\beta+v^3h_\gamma+uvh_\delta+uv^2h_{\delta'})\hat{f}_i+\langle x^n-1\rangle\rangle$ and $\langle(uv^2+v^3h_\alpha)\hat{f}_i+\langle x^n-1\rangle\rangle$ for $1\le i\le m$. Without loss of generality, we can assume that

$$\begin{aligned}C=&\langle\hat{f}_{m_1+1}\rangle\oplus\cdots\oplus\langle\hat{f}_{m_1+m_2}\rangle\oplus\langle v\hat{f}_{m_1+m_2+1}\rangle\oplus\cdots\oplus\langle v\hat{f}_{m_1+m_2+m_3}\rangle\oplus\langle v^2\hat{f}_{m_1+m_2+m_3+1}\rangle\\&\oplus\cdots\oplus\langle v\hat{f}_{m_1+m_2+m_3+m_4}\rangle\oplus\langle v^3\hat{f}_{m_1+m_2+m_3+m_4+1}\rangle\oplus\cdots\oplus\langle v^3\hat{f}_{m_1+m_2+m_3+m_4+m_5}\rangle\\&\oplus\langle uv\hat{f}_{m_1+m_2+m_3+m_4+m_5+1}\rangle\oplus\cdots\oplus\langle uv\hat{f}_{m_1+m_2+m_3+m_4+m_5+m_6}\rangle\\&\oplus\langle uv^2\hat{f}_{m_1+m_2+m_3+m_4+m_5+m_6+1}\rangle\oplus\cdots\oplus\langle uv^2\hat{f}_{m_1+m_2+m_3+m_4+m_5+m_6+m_7}\rangle\\&\oplus\langle uv^3\hat{f}_{m_1+\cdots+m_7+1}\rangle\oplus\cdots\oplus\langle uv^3\hat{f}_{m_1+\cdots+m_8}\rangle\oplus\langle u\hat{f}_{m_1+\cdots+m_8+1}\rangle\oplus\cdots\oplus\langle u\hat{f}_{m_1+\cdots+m_9}\rangle\\&\oplus(\langle uv^2\hat{f}_{m_1+\cdots+m_9+1}\rangle+\langle v^3\hat{f}_{m_1+\cdots+m_9+1}\rangle)\oplus\cdots\oplus(\langle uv^2\hat{f}_{m_1+\cdots+m_{10}}\rangle+\langle v^3\hat{f}_{m_1+\cdots+m_{10}}\rangle)\\&\oplus\langle(uv+v^2h_\alpha+v^3h_\beta+uv^2h_\delta)\hat{f}_{m_1+\cdots+m_{10}+1}\rangle\oplus\cdots\oplus\langle(uv+v^2h_\alpha+v^3h_\beta+uv^2h_\delta)\\&\hat{f}_{m_1+\cdots+m_{11}}\rangle\oplus\langle(u+vh_\alpha+v^2h_\beta+v^3h_\gamma+uvh_\delta+uv^2h_{\delta'})\hat{f}_{m_1+\cdots+m_{11}+1}\rangle\oplus\cdots\oplus\\&\langle(u+vh_\alpha+v^2h_\beta+v^3h_\gamma+uvh_\delta+uv^2h_{\delta'})\hat{f}_{m_1+\cdots+m_{12}}\rangle\oplus\langle(uv^2+v^3h_\alpha)\hat{f}_{m_1+\cdots+m_{12}+1}\rangle\\&\oplus\cdots\oplus\langle(uv^2+v^3h_\alpha)\hat{f}_m\rangle\end{aligned}$$

where integers $m_1, m_2, m_3, m_4, m_5, m_6, m_7, m_8, m_9, m_{10}, m_{11}, m_{12} \geq 0$ and $m_1 + \cdots + m_{12} + 1 \leq m$.

Let $m_0 = 0$ and $m_{13}$ be a non-negative integer such that $m_1 + \cdots + m_{13} = m$. Now, we define

$$\begin{aligned}
G_0 =& f_{m_0+1} \cdots f_{m_0+m_1}, G_1 = f_{m_0+m_1+1} \cdots f_{m_0+m_1+m_2},\\
G_2 =& f_{m_0+m_1+m_2+1} \cdots f_{m_0+m_1+m_2+m_3}, G_3 = f_{m_0+\cdots+m_3+1} \cdots f_{m_0+\cdots+m_4},\\
G_4 =& f_{m_0+\cdots+m_4+1} \cdots f_{m_0+\cdots+m_5}, G_5 = f_{m_0+\cdots+m_5+1} \cdots f_{m_0+\cdots+m_6},\\
G_6 =& f_{m_0+\cdots+m_6+1} \cdots f_{m_0+\cdots+m_7}, G_7 = f_{m_0+\cdots+m_7+1} \cdots f_{m_0+\cdots+m_8},\\
G_8 =& f_{m_0+\cdots+m_8+1} \cdots f_{m_0+\cdots+m_9}, G_9 = f_{m_0+\cdots+m_9+1} \cdots f_{m_0+\cdots+m_{10}},\\
G_{10} =& f_{m_0+\cdots+m_{10}+1} \cdots f_{m_0+\cdots+m_{11}}, G_{11} = f_{m_0+\cdots+m_{11}+1} \cdots f_{m_0+\cdots+m_{12}},\\
G_{12} =& f_{m_0+\cdots+m_{12}+1} \cdots f_m.
\end{aligned}$$

Then, from above construction, $G_0, G_1, \ldots, G_{12}$ are pairwise coprime and $\prod_{i=0}^{12} G_i = x^n - 1$. Also,

$$\begin{aligned}
C =& \langle \hat{G_1} \rangle \oplus \langle v\hat{G_2} \rangle \oplus \langle v^2\hat{G_3} \rangle \oplus \langle v^3\hat{G_4} \rangle \oplus \langle uv\hat{G_5} \rangle \oplus \langle uv^2\hat{G_6} \rangle \oplus \langle uv^3\hat{G_7} \rangle \oplus \langle u\hat{G_8} \rangle \oplus (\langle uv^2\hat{G_9} \rangle\\
&+ \langle v^3\hat{G_9} \rangle) \oplus \langle (uv + v^2h_\alpha + v^3h_\beta + uv^2h_\delta)\hat{G_{10}} \rangle \oplus \langle (u + vh_\alpha + v^2h_\beta + v^3h_\gamma + uvh_\delta\\
&+ uv^2h_{\delta'})\hat{G_{11}} \rangle \oplus \langle (uv^2 + v^3h_\alpha)\hat{G_{12}} \rangle.
\end{aligned}$$

Furthermore, $|\langle \hat{G_1} \rangle| = 16^{8\deg(G_1)}$, $|\langle v\hat{G_2} \rangle| = 16^{6\deg(G_2)}$, $|\langle v^2\hat{G_3} \rangle| = 16^{4\deg(G_3)}$, $|\langle v^3\hat{G_4} \rangle| = 16^{2\deg(G_4)}, |\langle uv\hat{G_5} \rangle| = 16^{3\deg(G_5)}, |\langle uv^2\hat{G_6} \rangle| = 16^{2\deg(G_6)}, |\langle uv^3\hat{G_7} \rangle| = 16^{\deg(G_7)}, |u\hat{G_8}| = 16^{4\deg(G_8)}$, $|(\langle uv\hat{G_9} \rangle + \langle v^3\hat{G_9} \rangle)| = 16^{3\deg(G_9)}$, $|\langle (uv + v^2h_\alpha + v^3h_\beta + uv^2h_\delta)\hat{G_{10}} \rangle| = 16^{3\deg(G_{10})}$, $|\langle (u + vh_\alpha + v^2h_\beta + v^3h_\gamma + uvh_\delta + uv^2h_{\delta'})\hat{G_{11}} \rangle| = 16^{5\deg(G_{11})}, |\langle (uv^2 + v^3h_\alpha)\hat{G_{12}} \rangle| = 16^{2\deg(G_{12})}$. Thus, $|C| = 16^{\xi}$, where $\xi = 8\deg(G_1) + 6\deg(G_2) + 4\deg(G_3) + 2\deg(G_4) + 3\deg(G_5) + 2\deg(G_6) + \deg(G_7) + 4\deg(G_8) + 3\deg(G_9) + 3\deg(G_{10}) + 5\deg(G_{11}) + 2\deg(G_{12})$.

Next, we consider $R' = \frac{\mathbb{F}_{16}[x]}{\langle x^n-1 \rangle_{\mathbb{F}_{16}[x]}}$ where $\langle x^n - 1 \rangle_{\mathbb{F}_{16}[x]}$ is an ideal of $\mathbb{F}_{16}[x]$ generated by $x^n - 1$.

**Theorem 4** *Let $C$ be a cyclic code of odd length $n$ over $R$. Then there exist polynomials $A, B, D, E, F, G, H, I, J, K, L$ in $\mathbb{F}_{16}[x]$ which are factors of $x^n - 1$ such that*

$$\begin{aligned}
C =& \langle A \rangle_{R'} + v\langle B \rangle_{R'} + v^2\langle D \rangle_{R'} + v^3\langle E \rangle_{R'} + u\langle F \rangle_{R'} + uv\langle G \rangle_{R'} + uv^2\langle H \rangle_{R'} + uv^3\langle I \rangle_{R'}\\
&+ (uv + v^2h_\alpha + v^3h_\beta + uv^2h_\delta)\langle J \rangle_{R'} + (u + vh_\alpha + v^2h_\beta + v^3h_\gamma + uvh_\delta + uv^2h_{\delta'})\langle K \rangle_{R'}\\
&+ (uv^2 + v^3h_\alpha)\langle L \rangle_{R'},
\end{aligned}$$

*where $h_\alpha, h_\beta, h_\gamma, h_\delta, h_{\delta'}$ are units in $\mathbb{F}_{16}[x]$ and $\langle - \rangle_{R'}$ is an ideal of $R'$ generated by $-$. Also,*

$$|C| = 16^{11n - (\deg A + \deg B + \deg D + \deg E + \deg F + \deg G + \deg H + \deg I + \deg J + \deg K + \deg L)}.$$

***Proof*** From Theorem 3, we have

$$\begin{aligned}
C =& \langle \hat{G_1} \rangle \oplus \langle v\hat{G_2} \rangle \oplus \langle v^2\hat{G_3} \rangle \oplus \langle v^3\hat{G_4} \rangle \oplus \langle uv\hat{G_5} \rangle \oplus \langle uv^2\hat{G_6} \rangle \oplus \langle uv^3\hat{G_7} \rangle \oplus \langle u\hat{G_8} \rangle \oplus (\langle uv^2\hat{G_9} \rangle\\
&+ \langle v^3\hat{G_9} \rangle) \oplus \langle (uv + v^2h_\alpha + v^3h_\beta + uv^2h_\delta)\hat{G_{10}} \rangle \oplus \langle (u + vh_\alpha + v^2h_\beta + v^3h_\gamma + uvh_\delta\\
&+ uv^2h_{\delta'})\hat{G_{11}} \rangle \oplus \langle (uv^2 + v^3h_\alpha)\hat{G_{12}} \rangle.
\end{aligned}$$

It can be easily seen that $\langle \hat{G}_i \rangle = \langle \hat{G}_i \rangle_{R'} + v\langle \hat{G}_i \rangle_{R'} + v^2\langle \hat{G}_i \rangle_{R'} + v^3\langle \hat{G}_i \rangle_{R'} + u\langle \hat{G}_i \rangle_{R'} + uv\langle \hat{G}_i \rangle_{R'} + uv^2\langle \hat{G}_i \rangle_{R'} + uv^3\langle \hat{G}_i \rangle_{R'}$ for $1 \le i \le 12$. Now,

$$\begin{aligned}
C =& \langle \hat{G}_1 \rangle_{R'} + v\langle \hat{G}_1 \rangle_{R'} + v^2\langle \hat{G}_1 \rangle_{R'} + v^3\langle \hat{G}_1 \rangle_{R'} + u\langle \hat{G}_1 \rangle_{R'} + uv\langle \hat{G}_1 \rangle_{R'} + uv^2\langle \hat{G}_1 \rangle_{R'} + uv^3\langle \hat{G}_1 \rangle_{R'} \\
&+ v\langle \hat{G}_2 \rangle_{R'} + v^2\langle \hat{G}_2 \rangle_{R'} + v^3\langle \hat{G}_2 \rangle_{R'} + uv\langle \hat{G}_2 \rangle_{R'} + uv^2\langle \hat{G}_2 \rangle_{R'} + uv^3\langle \hat{G}_2 \rangle_{R'} + v^2\langle \hat{G}_3 \rangle_{R'} + v^3\langle \hat{G}_3 \rangle_{R'} \\
&+ uv^2\langle \hat{G}_3 \rangle_{R'} + uv^3\langle \hat{G}_3 \rangle_{R'} + v^3\langle \hat{G}_4 \rangle_{R'} + uv^3\langle \hat{G}_4 \rangle_{R'} + uv\langle \hat{G}_5 \rangle_{R'} + uv^2\langle \hat{G}_5 \rangle_{R'} + uv^3\langle \hat{G}_5 \rangle_{R'} \\
&+ uv^2\langle \hat{G}_6 \rangle_{R'} + uv^3\langle \hat{G}_6 \rangle_{R'} + uv^3\langle \hat{G}_7 \rangle_{R'} + u\langle \hat{G}_8 \rangle_{R'} + uv\langle \hat{G}_8 \rangle_{R'} + uv^2\langle \hat{G}_8 \rangle_{R'} + uv^3\langle \hat{G}_8 \rangle_{R'} \\
&+ uv^2\langle \hat{G}_9 \rangle_{R'} + uv^3\langle \hat{G}_9 \rangle_{R'} + v^3\langle \hat{G}_9 \rangle_{R'} + (uv + v^2h_\alpha + v^3h_\beta + uv^2h_\delta)\langle \hat{G_{10}} \rangle + v^3\langle \hat{G_{10}} \rangle \\
&+ uv^2\langle \hat{G_{10}} \rangle + uv^3\langle \hat{G_{10}} \rangle + (u + vh_\alpha + v^2h_\beta + v^3h_\gamma + uvh_\delta + uv^2h_{\delta'})\langle \hat{G_{11}} \rangle + v^2\langle \hat{G_{11}} \rangle \\
&+ v^3\langle \hat{G_{11}} \rangle + uv\langle \hat{G_{11}} \rangle + uv^2\langle \hat{G_{11}} \rangle + uv^3\langle \hat{G_{11}} \rangle + (uv^2 + v^3h_{\alpha'})\langle \hat{G_{12}} \rangle + uv^3\langle \hat{G_{12}} \rangle \\
=& \langle \hat{G}_1 \rangle_{R'} + v(\langle \hat{G}_1 \rangle_{R'} + \langle \hat{G}_2 \rangle_{R'}) + v^2(\langle \hat{G}_1 \rangle_{R'} + \langle \hat{G}_2 \rangle_{R'} + \langle \hat{G}_3 \rangle_{R'} + \langle \hat{G_{11}} \rangle_{R'}) + v^3(\langle \hat{G}_1 \rangle_{R'} + \langle \hat{G}_2 \rangle_{R'} \\
&+ \langle \hat{G}_3 \rangle_{R'} + \langle \hat{G}_4 \rangle_{R'} + \langle \hat{G}_9 \rangle_{R'} + \langle \hat{G_{10}} \rangle_{R'} + \langle \hat{G_{11}} \rangle_{R'}) + u(\langle \hat{G}_1 \rangle_{R'} + \langle \hat{G}_8 \rangle_{R'}) + uv(\langle \hat{G}_1 \rangle_{R'} + \langle \hat{G}_2 \rangle_{R'} \\
&+ \langle \hat{G}_5 \rangle_{R'} + \langle \hat{G}_8 \rangle_{R'} + \langle \hat{G_{11}} \rangle_{R'}) + uv^2(\langle \hat{G}_1 \rangle_{R'} + \langle \hat{G}_2 \rangle_{R'} + \langle \hat{G}_3 \rangle_{R'} + \langle \hat{G}_5 \rangle_{R'} + \langle \hat{G}_6 \rangle_{R'} + \langle \hat{G}_8 \rangle_{R'} \\
&+ \langle \hat{G}_9 \rangle_{R'} + \langle \hat{G_{10}} \rangle_{R'} + \langle \hat{G_{11}} \rangle_{R'}) + uv^3(\langle \hat{G}_1 \rangle_{R'} + \langle \hat{G}_2 \rangle_{R'} + \langle \hat{G}_3 \rangle_{R'} + \langle \hat{G}_4 \rangle_{R'} + \langle \hat{G}_5 \rangle_{R'} + \langle \hat{G}_6 \rangle_{R'} \\
&+ \langle \hat{G}_7 \rangle_{R'} + \langle \hat{G}_8 \rangle_{R'} + \langle \hat{G}_9 \rangle_{R'} + \langle \hat{G_{10}} \rangle_{R'} + \langle \hat{G_{11}} \rangle_{R'} + \langle \hat{G_{12}} \rangle_{R'}) + (uv + v^2h_\alpha + v^3h_\beta + uv^2h_\delta)\langle \hat{G_{10}} \rangle \\
&+ (u + vh_\alpha + v^2h_\beta + v^3h_\gamma + uvh_\delta + uv^2h_{\delta'})\langle \hat{G_{11}} \rangle + (uv^2 + v^3h_\alpha)\langle \hat{G_{12}} \rangle.
\end{aligned}$$

Consider

$$\begin{aligned}
A =& \hat{G_1};\ B = \hat{G_1} + \hat{G_2};\ D = \hat{G_1} + \hat{G_2} + \hat{G_3} + \hat{G_{11}};\\
E =& \hat{G_1} + \hat{G_2} + \hat{G_3} + \hat{G_4} + \hat{G_9} + \hat{G_{10}} + \hat{G_{11}};\\
F =& \hat{G_1} + \hat{G_8};\ G = \hat{G_1} + \hat{G_2} + \hat{G_5} + \hat{G_8} + \hat{G_{11}};\\
H =& \hat{G_1} + \hat{G_2} + \hat{G_3} + \hat{G_5} + \hat{G_6} + \hat{G_8} + \hat{G_9} + \hat{G_{10}} + \hat{G_{11}};\\
I =& \hat{G_1} + \hat{G_2} + \hat{G_3} + \hat{G_4} + \hat{G_5} + \hat{G_6} + \hat{G_7} + \hat{G_8} + \hat{G_9} + \hat{G_{10}} + \hat{G_{11}} + \hat{G_{12}};\\
J =& \hat{G_{10}};\ K = \hat{G_{11}};\ L = \hat{G_{12}}.
\end{aligned}$$

Now, we have to prove $\langle \hat{B} \rangle_{R'} = \langle \hat{G_1} \rangle_{R'} + \langle \hat{G_2} \rangle_{R'}$, $\langle \hat{D} \rangle_{R'} = \langle \hat{G_1} \rangle_{R'} + \langle \hat{G_2} \rangle_{R'} + \langle \hat{G_3} \rangle_{R'} + \langle \hat{G_{11}} \rangle_{R'}$, $\langle \hat{E} \rangle_{R'} = \langle \hat{G_1} \rangle_{R'} + \langle \hat{G_2} \rangle_{R'} + \langle \hat{G_3} \rangle_{R'} + \langle \hat{G_4} \rangle_{R'} + \langle \hat{G_9} \rangle_{R'} + \langle \hat{G_{10}} \rangle_{R'} + \langle \hat{G_{11}} \rangle_{R'}$, $\langle \hat{F} \rangle_{R'} = \langle \hat{G_1} \rangle_{R'} + \langle \hat{G_8} \rangle_{R'}$, $\langle \hat{G} \rangle_{R'} = \langle \hat{G_1} \rangle_{R'} + \langle \hat{G_2} \rangle_{R'} + \langle \hat{G_5} \rangle_{R'} + \langle \hat{G_8} \rangle_{R'} + \langle \hat{G_{11}} \rangle_{R'}$, $\langle \hat{H} \rangle_{R'} = \langle \hat{G_1} \rangle_{R'} + \langle \hat{G_2} \rangle_{R'} + \langle \hat{G_3} \rangle_{R'} + \langle \hat{G_5} \rangle_{R'} + \langle \hat{G_6} \rangle_{R'} + \langle \hat{G_8} \rangle_{R'} + \langle \hat{G_9} \rangle_{R'} + \langle \hat{G_{10}} \rangle_{R'} + \langle \hat{G_{11}} \rangle_{R'}$, $\langle \hat{I} \rangle_{R'} = \langle \hat{G_1} \rangle_{R'} + \langle \hat{G_2} \rangle_{R'} + \langle \hat{G_3} \rangle_{R'} + \langle \hat{G_4} \rangle_{R'} + \langle \hat{G_5} \rangle_{R'} + \langle \hat{G_6} \rangle_{R'} + \langle \hat{G_7} \rangle_{R'} + \langle \hat{G_8} \rangle_{R'} + \langle \hat{G_9} \rangle_{R'} + \langle \hat{G_{10}} \rangle_{R'} + \langle \hat{G_{11}} \rangle_{R'} + \langle \hat{G_{12}} \rangle_{R'}$.

For any distinct $i, j \in \{0, 1, \ldots, 12\}$, $x^n - 1 | \hat{G}_i \hat{G}_j$ and $G_i G_j = 0$ in $\frac{R[x]}{\langle x^n - 1 \rangle}$. Since $\{G_i, \hat{G}_i\}$ are coprime pairs for $i = 1, \ldots, 12$, there exist $a_{0i}, a_{1i} \in \mathbb{F}_{16}[x]$ such that $a_{0i}G_i + a_{1i}\hat{G}_i = 1$.

Further, we consider $a_{0i}G_i + a_{1i}\hat{G}_i = 1$, for $i = 2, \ldots, 12$, then there exist polynomials $w_1, w_2, \cdots, w_{12}$ in $\mathbb{F}_{16}[x]$ such that

$$w_1 G_2 \cdots G_{12} + w_2 \hat{G}_2 G_3 \cdots G_{12} + \cdots + w_{12} G_2 G_3 \cdots G_{11} \hat{G_{12}} = 1.$$

Multiplying above equation by $\hat{G}_1$, we have $\hat{G}_1 = w_1 \hat{G}_1 G_2 \cdots G_{12}$. By the above construction, $B = \hat{G}_1 + \hat{G}_2$. This implies that $w_1 A G_2 \cdots G_{12} = w_1 \hat{G}_1 G_2 \cdots G_{12} = \hat{G}_1$. Therefore, $w_1 B G_2 \cdots G_{12} = \hat{G}_1$. Hence, we conclude that $\hat{G}_1 \in \langle B \rangle_{R'}$. Continuing in this manner, we have $\hat{G}_1, \hat{G}_2 \in \langle B \rangle_{R'}$. Similarly, we can prove $\hat{G}_1, \hat{G}_2, \hat{G}_3, \hat{G_{11}} \in \langle D \rangle_{R'}$, $\hat{G}_1, \hat{G}_2, \hat{G}_3, \hat{G}_4, \hat{G}_9, \hat{G_{10}}, \hat{G_{11}} \in \langle E \rangle_{R'}$, $\hat{G}_1, \hat{G}_8 \in \langle F \rangle_{R'}$, $\hat{G}_1, \hat{G}_2, \hat{G}_5, \hat{G}_8, \hat{G_{11}} \in \langle G \rangle_{R'}$, $\hat{G}_1, \hat{G}_2, \hat{G}_3, \hat{G}_5, \hat{G}_6, \hat{G}_8, \hat{G}_9, \hat{G_{10}}, \hat{G_{11}} \in \langle H \rangle_{R'}$, $\hat{G}_1, \hat{G}_2, \hat{G}_3, \hat{G}_4, \hat{G}_5, \hat{G}_6, \hat{G}_7, \hat{G}_8, \hat{G}_9$, $\hat{G_{10}}, \hat{G_{11}}, \hat{G_{12}} \in \langle I \rangle_{R'}$. Hence,

$$C = \langle A\rangle_{R'} + v\langle B\rangle_{R'} + v^2\langle D\rangle_{R'} + v^3\langle E\rangle_{R'} + u\langle F\rangle_{R'} + uv\langle G\rangle_{R'} + uv^2\langle H\rangle_{R'} + uv^3\langle I\rangle_{R'} + (uv + v^2h_\alpha + v^3h_\beta + uv^2h_\delta)\langle J\rangle_{R'} + (u + vh_\alpha + v^2h_\beta + v^3h_\gamma + uvh_\delta + uv^2h_{\delta'})\langle K\rangle_{R'} + (uv^2 + v^3h_\alpha)\langle L\rangle_{R'}.$$

Since, $|\mathbb{F}_{16}| = 16, |\langle A\rangle_{R'}| = 16^{n-\deg A}$ and hence

$$|C| = 16^{11n-(\deg A+\deg B+\deg D+\deg E+\deg F+\deg G+\deg H+\deg I+\deg J+\deg K+\deg L)}.$$

## 3 Euclidean dual of cyclic codes over $M_4(\mathbb{F}_2 + u\mathbb{F}_2)$

For given $a = (a_0, a_1, \ldots, a_{n-1})$ and $b = (b_0, b_1, \ldots, b_{n-1})$ in $R^n$, the *Euclidean inner product* of $a$ and $b$ is defined as $a \cdot b = a_0b_0 + a_1b_1 + \cdots + a_{n-1}b_{n-1}$. Two vectors $a$ and $b$ are said to be orthogonal if $a \cdot b = 0$. For a linear code $C$ over $R$, its dual $C^\perp$ is defined as $C^\perp = \{a \in C \mid a \cdot b = 0\ \forall\ b \in C\}$. A code $C$ is called self-orthogonal if $C \subseteq C^\perp$ and self-dual if $C = C^\perp$. For each polynomial $f(x) = f_0 + f_1x + \cdots + f_{n-1}x^{n-1}$ with $f_{n-1} \neq 0$, the reciprocal of $f(x)$ is defined as $f^*(x) = x^{n-1}f(1/x) = f_{n-1} + f_{n-2}x + \cdots + f_0x^{n-1}$. Note that $\deg f^*(x) \leq \deg f(x)$, and if $f_0 \neq 0$, then $\deg f^*(x) = \deg f(x)$.

**Theorem 5** *Let $C$ be a cyclic code of odd length $n$ over $R$ with* $C = \langle \hat{G_1}\rangle \oplus \langle v\hat{G_2}\rangle \oplus \langle v^2\hat{G_3}\rangle \oplus\langle v^3\hat{G_4}\rangle \oplus \langle uv\hat{G_5}\rangle \oplus \langle uv^2\hat{G_6}\rangle \oplus \langle uv^3\hat{G_7}\rangle \oplus \langle u\hat{G_8}\rangle \oplus (\langle uv^2\hat{G_9}\rangle + \langle v^3\hat{G_9}\rangle) \oplus \langle (uv + v^2h_\alpha + v^3h_\beta + uv^2h_\delta)\hat{G_{10}}\rangle \oplus \langle (u + vh_\alpha + v^2h_\beta + v^3h_\gamma + uvh_\delta + uv^2h_{\delta'})\hat{G_{11}}\rangle \oplus \langle (uv^2 + v^3h_\alpha)\hat{G_{12}}\rangle$ *where $h_\alpha, h_\beta, h_\gamma, h_\delta, h_{\delta'}$ are units in $\mathbb{F}_{16}[x]$. Then*

$$C^\perp = \langle \hat{G_0^*}\rangle \oplus \langle v^3\hat{G_2^*}\rangle \oplus \langle v^2\hat{G_3^*}\rangle \oplus \langle v\hat{G_4^*}\rangle \oplus \langle (u + vh_\alpha + v^2h_\beta + v^3h_\gamma + uvh_\delta + uv^2h_{\delta'})\hat{G_5^*}\rangle \oplus (\langle u\hat{G_6^*}\rangle + \langle v^2\hat{G_6^*}\rangle) \oplus (\langle u\hat{G_7^*}\rangle + \langle v\hat{G_7^*}\rangle) \oplus \langle u\hat{G_8^*}\rangle \oplus (\langle (uv + v^2h_\alpha + v^3h_\beta + uv^2h_\delta)\hat{G_9^*}\rangle + \langle v^3\hat{G_9^*}\rangle) \oplus (\langle (uv + v^2h_\alpha + v^3h_\beta + uv^2h_\delta)\hat{G_{10}^*}\rangle + \langle uv^2\hat{G_{10}^*}\rangle + \langle v^3\hat{G_{10}^*}\rangle) \oplus (\langle uv^3\hat{G_{11}^*}\rangle + \langle v^2\hat{G_{11}^*}\rangle) \oplus (\langle (u + vh_\alpha + v^2h_\beta + v^3h_\gamma + uvh_\delta + uv^2h_{\delta'})\hat{G_{12}^*}\rangle + \langle v^3\hat{G_{12}^*}\rangle),$$

*and* $|C^\perp| = 16^\eta$, *where* $\eta = 8\deg(G_0) + 2\deg(G_2) + 4\deg(G_3) + 6\deg(G_4) + 5\deg(G_5) + 6\deg(G_6) + 7\deg(G_7) + 4\deg(G_8) + 5\deg(G_9) + 5\deg(G_{10}) + 3\deg(G_{11}) + 6\deg(G_{12})$.

***Proof*** Let $C$ be a cyclic code of odd length $n$ over $R$. From Theorem 3, $|C| = 16^\xi$ where $\xi = 8\deg(G_1) + 6\deg(G_2) + 4\deg(G_3) + 2\deg(G_4) + 3\deg(G_5) + 2\deg(G_6) + \deg(G_7) + 4\deg(G_8) + 3\deg(G_9) + 3\deg(G_{10}) + 5\deg(G_{11}) + 2\deg(G_{12})$. Since$|C||C^\perp| = 16^{8n}$ and$n = \deg(G_0) + \deg(G_1) + \deg(G_2) + \deg(G_3) + \deg(G_4) + \deg(G_5) + \deg(G_6) + \deg(G_7) + \deg(G_8) + \deg(G_8) + \deg(G_9) + \deg(G_{10}) + \deg(G_{11}) + \deg(G_{12})$ therefore, $|C^\perp| = 16^\eta$

where $\eta = 8n - \xi = 8\deg(G_0) + 2\deg(G_2) + 4\deg(G_3) + 6\deg(G_4) + 5\deg(G_5) + 6\deg(G_6) + 7\deg(G_7) + 4\deg(G_8) + 5\deg(G_9) + 5\deg(G_{10}) + 3\deg(G_{11}) + 6\deg(G_{12})$.

Now, consider $C^* = \langle \hat{G_0^*}\rangle \oplus \langle v^3\hat{G_2^*}\rangle \oplus \langle v^2\hat{G_3^*}\rangle \oplus \langle v\hat{G_4^*}\rangle \oplus \langle (u + vh_\alpha + v^2h_\beta + v^3h_\gamma + uvh_\delta + uv^2h_{\delta'})\hat{G_5^*}\rangle \oplus (\langle u\hat{G_6^*}\rangle + \langle v^2\hat{G_6^*}\rangle) \oplus (\langle u\hat{G_7^*}\rangle + \langle v\hat{G_7^*}\rangle) \oplus \langle u\hat{G_8^*}\rangle \oplus (\langle (uv + v^2h_\alpha + v^3h_\beta + uv^2h_\delta)\hat{G_{10}^*}\rangle + \langle uv^2\hat{G_{10}^*}\rangle + \langle v^3\hat{G_{10}^*}\rangle) \oplus (\langle uv^3\hat{G_{11}^*}\rangle + \langle v^2\hat{G_{11}^*}\rangle) \oplus (\langle (u + vh_\alpha + v^2h_\beta + v^3h_\gamma + uvh_\delta$

$+uv^2h_{\delta'})\hat{G^*_{12}}\rangle + \langle v^3\hat{G^*_{12}}\rangle)$. For $i$, $j$ and $0 \le i,j \le 12$, if $i+1 = 12-j+1$, it can be easily seen that

$$v^4\hat{G}_2\hat{G}_2^* = 0, v^4\hat{G}_3\hat{G}_3^* = 0, v^4\hat{G}_4\hat{G}_4^* = 0,$$
$$uv(u+vh_\alpha+v^2h_\beta+v^3h_\gamma+uvh_\delta+uv^2h_{\delta'})\hat{G}_5\hat{G}_5^* = 0,$$
$$uv^2(\langle u\rangle+\langle v^2\rangle)\hat{G}_6\hat{G}_6^* = 0,\ uv^3(\langle u\rangle+\langle v\rangle)\hat{G}_7\hat{G}_7^* = 0,\ u^2\hat{G}_8\hat{G}_8^* = 0,$$
$$(\langle uv^2\rangle+\langle v^3\rangle)(\langle(uv+v^2h_\alpha+v^3h_\beta+uv^2h_\delta)\rangle+\langle v^3\rangle)\hat{G}_9\hat{G}_9^* = 0,$$
$$(uv+v^2h_\alpha+v^3h_\beta+uv^2h_\delta)(\langle(uv+v^2h_\alpha+v^3h_\beta+uv^2h_\delta)\rangle+\langle uv^2\rangle+\langle v^3\rangle)\hat{G_{10}}\hat{G^*_{10}} = 0,$$
$$(u+vh_\alpha+v^2h_\beta+v^3h_\gamma+uvh_\delta+uv^2h_{\delta'})(\langle uv^3\rangle+\langle v^2\rangle)\hat{G_{11}}\hat{G^*_{11}} = 0,$$
$$(uv^2+v^3h_\alpha)(\langle(u+vh_\alpha+v^2h_\beta+v^3h_\gamma+uvh_\delta+uv^2h_{\delta'})\rangle+\langle v^3\rangle)\hat{G_{12}}\hat{G^*_{12}} = 0.$$

Precisely, we can say that $\hat{G}_{i+1}\hat{G}^*_{12-j+1} = 0$.

If $i+1 \neq 12-j+1$, i.e., $i \neq 12-j$ then $x^n-1|\hat{G}_{i+1}\hat{G}^*_{12-j+1}$. This implies that $\hat{G}_{i+1}\hat{G}^*_{12-j+1} = 0$. Hence, $C^* \subseteq C^\perp$.

Furthermore, $|\hat{G}_0^*| = 16^{8\deg(G_0)}$, $|v^3\hat{G}_2^*| = 16^{2\deg(G_2)}$, $|v^2\hat{G}_3^*| = 16^{4\deg(G_3)}$, $|v\hat{G}_4^*| = 16^{6\deg(G_4)}$, $|(u+vh_\alpha+v^2h_\beta+v^3h_\gamma+uvh_\delta+uv^2h_{\delta'})\hat{G}_5^*| = 16^{5\deg(G_5)}$, $|(\langle u\hat{G}_6^*\rangle+\langle v^2\hat{G}_6^*\rangle)| = 16^{6\deg(G_6)}$, $|(\langle u\hat{G}_7^*\rangle+\langle v\hat{G}_7^*\rangle)| = 16^{7\deg(G_7)}$, $|u\hat{G}_8^*| = 16^{4\deg(G_8)}$, $|(\langle(uv+v^2h_\alpha+v^3h_\beta+uv^2h_\delta)\hat{G}_9^*\rangle+\langle v^3\hat{G}_9^*\rangle)| = 16^{5\deg(G_9)}$, $|(\langle(uv+v^2h_\alpha+v^3h_\beta+uv^2h_\delta)\hat{G^*_{10}}\rangle+\langle uv^2\hat{G^*_{10}}\rangle+\langle v^3\hat{G^*_{10}}\rangle)| = 16^{5\deg(G_{10})}$, $|(\langle uv^3\hat{G^*_{11}}\rangle+\langle v^2\hat{G^*_{11}}\rangle)| = 16^{3\deg(G_{11})}$, $|(\langle(u+vh_\alpha+v^2h_\beta+v^3h_\gamma+uvh_\delta+uv^2h_{\delta'})\hat{G^*_{12}}\rangle+\langle v^3\hat{G^*_{12}}\rangle)| = 16^{6\deg(G_{12})}$.

Therefore, $|C^*| = 16^\eta = |C^\perp|$ where $\eta = 8\deg(G_0)+2\deg(G_2)+4\deg(G_3)+6\deg(G_4)+5\deg(G_5)+6\deg(G_6)+7\deg(G_7)+4\deg(G_8)+5\deg(G_9)+5\deg(G_{10})+3\deg(G_{11})+6\deg(G_{12})$ Consequently, we get $C^* = C^\perp$.

**Theorem 6** *Let $C$ be a cyclic code of odd length $n$ over $R$. Then there exist polynomials $A^*, B^*, D^*, E^*, F^*, G^*, H^*, I^*, J^*, K^*, L^*$ in $\mathbb{F}_{16}[x]$ which are factors of $x^n-1$ such that*

$$\begin{aligned}C^\perp =&\langle A^*\rangle_{R'}+v\langle B^*\rangle_{R'}+v^2\langle D^*\rangle_{R'}+v^3\langle E^*\rangle_{R'}+u\langle F^*\rangle_{R'}+uv\langle G^*\rangle_{R'}+uv^2\langle H^*\rangle_{R'}+uv^3\langle I^*\rangle_{R'}\\&+(uv+v^2h_\alpha+v^3h_\beta+uv^2h_\delta)\langle J^*\rangle_{R'}+(u+vh_\alpha+v^2h_\beta+v^3h_\gamma+uvh_\delta+uv^2h_{\delta'})\langle K^*\rangle_{R'}\\&+(uv^2+v^3h_\alpha)\langle L^*\rangle_{R'},\end{aligned}$$

*where $h_\alpha, h_\beta, h_\gamma, h_\delta, h_{\delta'}$ are units in $\mathbb{F}_{16}[x]$ and $\langle-\rangle_{R'}$ is an ideal of $R'$ generated by $-$. Moreover,*

$$|C^\perp| = 16^{11n-(\deg A^*+\deg B^*+\deg D^*+\deg E^*+\deg F^*+\deg G^*+\deg H^*+\deg I^*+\deg J^*+\deg K^*+\deg L^*)}.$$

***Proof*** Follows from Theorems 4 and 5.

## 4 Hermitian dual of cyclic codes over $M_4(\mathbb{F}_2+u\mathbb{F}_2)$

For any two codewords $a = (a_0, a_1, \ldots, a_{n-1})$ and $b = (b_0, b_1, \ldots, b_{n-1})$ in $R^n$, the *Hermitian inner product* of $a$ and $b$ is defined as

$$< a, b >= a_0\overline{b_0} + a_1\overline{b_1} + \cdots + a_{n-1}\overline{b_{n-1}},$$

where "−" stands for the conjugation and $\overline{b_i} = b_i^4$ for $i = 0, 1, \ldots, n-1$. The Hermitian dual $C^{\perp H}$ of $C$ is defined as $C^{\perp H} = \{a \in C \mid < a, b >= 0 \ \forall\ b \in C\}$. It can be easily seen that $\overline{C^{\perp}} = C^{\perp H}$. A code $C$ is called Hermitian self-orthogonal if $C \subseteq C^{\perp H}$ and Hermitian self-dual if $C = C^{\perp H}$. For each polynomial $f(x) = f_0 + f_1 x + \cdots + f_k x^k$ with $f_k \neq 0$, the reciprocal of $f(x)$ is defined as $f^*(x) = x^k f(1/x) = f_k + f_{k-1}x + \cdots + f_0 x^k$. Note that $\deg f^*(x) \leq \deg f(x)$, and if $f_0 \neq 0$, then $\deg f^*(x) = \deg f(x)$. We denote $\overline{f(x)} = f_0^4 + f_1^4 x + \cdots + f_k^4 x^k$ and also it is easy to verify $\overline{f^*(x)} = \overline{f(x)}^*$.

**Theorem 7** *Let $C$ be a cyclic code of odd length $n$ over $R$ with* $C = \langle \hat{G_1} \rangle \oplus \langle v\hat{G_2} \rangle \oplus \langle v^2\hat{G_3} \rangle \oplus \langle v^3\hat{G_4} \rangle \oplus \langle uv\hat{G_5} \rangle \oplus \langle uv^2\hat{G_6} \rangle \oplus \langle uv^3\hat{G_7} \rangle \oplus \langle u\hat{G_8} \rangle \oplus (\langle uv^2\hat{G_9} \rangle + \langle v^3\hat{G_9} \rangle) \oplus \langle (uv + v^2h_\alpha + v^3h_\beta + uv^2h_\delta)\hat{G_{10}} \rangle \oplus \langle (u + vh_\alpha + v^2h_\beta + v^3h_\gamma + uvh_\delta + uv^2h_{\delta'})\hat{G_{11}} \rangle \oplus \langle (uv^2 + v^3h_\alpha)\hat{G_{12}} \rangle$ *where $h_\alpha, h_\beta, h_\gamma, h_\delta, h_{\delta'}$ are units in $\mathbb{F}_{16}[x]$. Then*

$$\begin{aligned} C^{\perp H} =& \langle \hat{\overline{G_0^*}} \rangle \oplus \langle v^3\hat{\overline{G_2^*}} \rangle \oplus \langle v^2\hat{\overline{G_3^*}} \rangle \oplus \langle v\hat{\overline{G_4^*}} \rangle \oplus \langle (u + vh_\alpha + v^2h_\beta + v^3h_\gamma + uvh_\delta + uv^2h_{\delta'})\hat{\overline{G_5^*}} \rangle \\ & \oplus (\langle u\hat{\overline{G_6^*}} \rangle + \langle v^2\hat{\overline{G_6^*}} \rangle) \oplus (\langle u\hat{\overline{G_7^*}} \rangle + \langle v\hat{\overline{G_7^*}} \rangle) \oplus \langle u\hat{\overline{G_8^*}} \rangle \oplus (\langle (uv + v^2h_\alpha + v^3h_\beta + uv^2h_\delta)\hat{\overline{G_9^*}} \rangle \\ & + \langle v^3\hat{\overline{G_9^*}} \rangle) \oplus (\langle (uv + v^2h_\alpha + v^3h_\beta + uv^2h_\delta)\hat{\overline{G_{10}^*}} \rangle + \langle uv^2\hat{\overline{G_{10}^*}} \rangle + \langle v^3\hat{\overline{G_{10}^*}} \rangle) \oplus (\langle uv^3\hat{\overline{G_{11}^*}} \rangle \\ & + \langle v^2\hat{\overline{G_{11}^*}} \rangle) \oplus (\langle (u + vh_\alpha + v^2h_\beta + v^3h_\gamma + uvh_\delta + uv^2h_{\delta'})\hat{\overline{G_{12}^*}} \rangle + \langle v^3\hat{\overline{G_{12}^*}} \rangle) \end{aligned}$$

*and* $|C^{\perp H}| = 16^\eta$ *where* $\eta = 8\deg(G_0) + 2\deg(G_2) + 4\deg(G_3) + 6\deg(G_4) + 5\deg(G_5) + 6\deg(G_6) + 7\deg(G_7) + 4\deg(G_8) + 5\deg(G_9) + 5\deg(G_{10}) + 3\deg(G_{11}) + 6\deg(G_{12})$

***Proof*** Follows the same procedure as the proof of Theorem 5.

**Theorem 8** *Let $C$ be a cyclic code of odd length $n$ over $R$. Then there exist polynomials $\overline{A^*}, \overline{B^*}, \overline{D^*}, \overline{E^*}, \overline{F^*}, \overline{G^*}, \overline{H^*}, \overline{I^*}, \overline{J^*}, \overline{K^*}, \overline{L^*}$ in $\mathbb{F}_{16}[x]$ which are factors of $x^n - 1$ such that*

$$\begin{aligned} C^{\perp H} =& \langle \overline{A^*} \rangle_{R'} + v\langle \overline{B^*} \rangle_{R'} + v^2\langle \overline{D^*} \rangle_{R'} + v^3\langle \overline{E^*} \rangle_{R'} + u\langle \overline{F^*} \rangle_{R'} + uv\langle \overline{G^*} \rangle_{R'} + uv^2\langle \overline{H^*} \rangle_{R'} \\ & + uv^3\langle \overline{I^*} \rangle_{R'} + (uv + v^2h_\alpha + v^3h_\beta + uv^2h_\delta)\langle \overline{J^*} \rangle_{R'} + (u + vh_\alpha + v^2h_\beta + v^3h_\gamma \\ & + uvh_\delta + uv^2h_{\delta'})\langle \overline{K^*} \rangle_{R'} + (uv^2 + v^3h_\alpha)\langle \overline{L^*} \rangle_{R'} \end{aligned}$$

*where $h_\alpha, h_\beta, h_\gamma, h_\delta, h_{\delta'}$ are units in $\mathbb{F}_{16}[x]$ and $\langle - \rangle_{R'}$ is an ideal of $R'$ generated by $-$. Moreover,*

$$|C^{\perp H}| = 16^{11n - (\deg A^* + \deg B^* + \deg D^* + \deg E^* + \deg F^* + \deg G^* + \deg H^* + \deg I^* + \deg J^* + \deg K^* + \deg L^*)}.$$

***Proof*** Using the similar arguments of Theorems 4 and 7, we get the required result.

## 5 Right $\mathbb{F}_2$-module isometry

### 5.1 The Gray map

We recall from [5], in case of ring $\mathbb{F}_2 + u\mathbb{F}_2$, the Lee weight is defined by $\omega_L(0) = 0$, $\omega_L(1) = \omega_L(1 + u) = 1$, $\omega_L(u) = 2$ and the Gray map from $(\mathbb{F}_2 + u\mathbb{F}_2)^n$ to $\mathbb{F}_2^{2n}$ sends $(a + ub)$ to $(b, a + b)$ for $a, b \in \mathbb{F}_2^n$. Now, it will help us to extend the Gray map $\theta$ from $R$ to $\mathbb{F}_{16}^8$ by

$$\begin{aligned}\theta(d_1 + vd_2 + v^2d_3 + v^3d_4 + ud_5 + uvd_6 + uv^2d_7 + uv^3d_8) =& (d_8, d_7 + d_8, d_6 + d_8, d_5 \\ &+ d_8, d_4 + d_8, d_3 + d_8, d_2 \\ &+ d_8, d_1 + d_2 + d_3 + d_4 \\ &+ d_5 + d_6 + d_7 + d_8)\end{aligned}$$

where $d_i \in \mathbb{F}_{16}$ for $i = 1, 2, \ldots, 8$. Also, $\theta$ is distance preserving. Therefore, for any element $r = d_1 + vd_2 + v^2d_3 + v^3d_4 + ud_5 + uvd_6 + uv^2d_7 + uv^3d_8$ in $R$, the extended Lee weight is defined as

$$\begin{aligned}\omega_L(r) =& \omega_H(d_8) + \omega_H(d_7 + d_8) + \omega_H(d_6 + d_8) + \omega_H(d_5 + d_8) + \omega_H(d_4 + d_8) + \omega_H(d_3 + d_8) \\ &+ \omega_H(d_2 + d_8) + \omega_H(d_1 + d_2 + d_3 + d_4 + d_5 + d_6 + d_7 + d_8)\end{aligned}$$

where $\omega_H(.)$ denotes the Hamming weight in $\mathbb{F}_{16}$. If $c = (c_0, c_1, \ldots, c_{n-1}) \in R^n$, then the Lee weight of $c$ is

$$\omega_L(c) = \omega_L(c_0) + \omega_L(c_1) + \cdots + \omega_L(c_{n-1}).$$

The Hamming weight $\omega_H(c)$ of a codeword $c = (c_0, c_1, \ldots, c_{n-1}) \in C$ is the number of non-zero components. For any two codewords $c$ and $c'$ of $C$, the Hamming distance is defined as $d_H(c, c') = \omega_H(c - c')$. The Hamming distance for the code $C$ is $d_H(C) = \min\{d_H(c, c') \mid c \neq c' \text{ for all } c, c' \in C\}$. The Lee distance between any two codewords $c$ and $c'$ is defined as $d_L(c, c') = \omega_L(c - c')$ and the Lee distance for the code $C$ is $d_L(C) = \min\{d_L(c, c') \mid c \neq c' \text{ for all } c, c' \in C\}$. Next, we have the following result.

**Theorem 9** *Let $C$ be a cyclic code over $R$ of length $n$, size $M$ and Lee distance $d$. Then $\theta(C)$ is a linear code over $\mathbb{F}_{16}$ of length $8n$, size $M$ and Hamming distance $d$.*

### 5.2 The Bachoc map

Here, we consider $R$ as a usual extension of the ring $M_4(\mathbb{F}_2)$ and extend the definition of Bachoc weight from $M_4(\mathbb{F}_2)$ to $R$ that was introduced in [3]. In [13], a theory of cyclic codes over $M_4(\mathbb{F}_2)$ was discussed by giving a characterization of cyclic codes and their duals as right ideals. But, these results were derived only for odd length. Now, by defining Bachoc map, we are able to construct codes of even length over $M_4(\mathbb{F}_2)$. Let $W_B$ be the Bachoc weight over $R$ and $w_B$ be the ordinary Bachoc weight of $M_4(\mathbb{F}_2)$ codes. For $Y_1, Y_2 \in M_4(\mathbb{F}_2)$, we define

$$W_B(Y = Y_1 + uY_2) = w_B(Y_2) + w_B(Y_1 + Y_2).$$

This immediately leads to a Bachoc map $\phi$ from $R$ to $M_4^2(\mathbb{F}_2)$ which is defined by

$$\phi(Y_1 + uY_2) = (Y_2, Y_1 + Y_2).$$

The map $\phi$ can be naturally extended to $R^n$ by preserving isometry as

$$\phi : (R^n, \text{Bachoc weight}) \to (M_4^{2n}(\mathbb{F}_2), \text{Bachoc weight}).$$

Now, we define the mapping $\psi$ as

$$\psi : M_4(\mathbb{F}_2 + u\mathbb{F}_2) \to \mathbb{F}_{16} + v\mathbb{F}_{16} + v^2\mathbb{F}_{16} + v^3\mathbb{F}_{16} + u\mathbb{F}_{16} + uv\mathbb{F}_{16} + uv^2\mathbb{F}_{16} + uv^3\mathbb{F}_{16}$$

$$\begin{pmatrix} a_1 + ub_1 & a_2 + ub_2 & a_3 + ub_3 & a_4 + ub_4 \\ (a_2 + a_5) + u(b_2 + b_5) & (a_3 + a_6) + u(b_3 + b_6) & (a_4 + a_7) + u(b_4 + b_7) & (a_1 + a_2 + a_3 + a_4 + a_8) \\ & & & +u(b_1 + b_2 + b_3 + b_4 + b_8) \\ a_9 + ub_9 & a_{10} + ub_{10} & a_{11} + ub_{11} & a_{12} + ub_{12} \\ (a_{10} + a_{13}) + u(b_{10} + b_{13}) & (a_{11} + a_{14}) + u(b_{11} + b_{14}) & (a_{12} + a_{15}) + u(b_{12} + b_{15}) & (a_9 + a_{10} + a_{11} + a_{12} + a_{16}) \\ & & & +u(b_9 + b_{10} + b_{11} + b_{12} + b_{16}) \end{pmatrix}$$

$$\to (a_1 + a_2w + a_3w^2 + a_4w^3) + v(b_1 + b_2w + b_3w^2 + b_4w^3) + v^2(a_5 + a_6w + a_7w^2 + a_8w^3)$$
$$+ v^3(b_5 + b_6w + b_7w^2 + b_8w^3) + u(a_9 + a_{10}w + a_{11}w^2 + a_{12}w^3) + uv(b_9 + b_{10}w + b_{11}w^2 + b_{12}w^3)$$
$$+ uv^2(a_{13} + a_{14}w + a_{15}w^2 + a_{16}w^3) + uv^3(b_{13} + b_{14}w + b_{15}w^2 + b_{16}w^3)$$

where $a_i, b_j \in \mathbb{F}_2$ for $1 \le i,j \le 16$. It can be easily seen that $\psi$ is a right $\mathbb{F}_2$-module isomorphism. Also,

$$W_B(Y = Y_1 + uY_2) = w_B(Y_2) + w_B(Y_1 + Y_2) = w_L(\psi(Y))$$

for all $Y \in R$, i.e., $\psi$ is a right-module isometry.

**Theorem 10** *Let $C$ be a cyclic code over $R$ of length $n$, size $M$ and Bachoc distance $d$. Then $\psi(C)$ is a linear code over $\mathbb{F}_{16}$ of length $8n$, size $M$ and Hamming distance $d$.*

## 6 Examples

It is well known that a $q$-ary linear code $[n, k, d]$ should satisfy the Singleton bound $d \le n - k + 1$, and said to be MDS (maximum-distance-separable) if $d = n - k + 1$. For a linear code $[n, k]$ , it is one of the fundamental problems to optimize $d$ and attain $d_{MDS} = n - k + 1$. Here, we call $[n, k, d]$ is a *good code* if $d_{MDS} - d \le 3$. This section contains many such good codes over $\mathbb{F}_{16}$ as the Gray images of cyclic codes where the parameters of the codes are calculated by using Magma Computation Software [6].

***Example 1*** For $n = 3$, we have $x^3 - 1 = (x + 1)(x + w^5)(x + w^{10}) \in \mathbb{F}_{16}[x]$. Let $f_1 = x + 1$, $f_2 = x + w^5$, $f_3 = x + w^{10}$. Then some good codes under the Gray images of cyclic codes of length 3 over $R$ are as follows.

| Generators of cyclic codes | Images under $\theta$ | $d_{MDS}$ | Remark |
|---|---|---|---|
| $\langle uf_2, vf_3, uvf_1, (uv^2 + v^3)f_1f_3\rangle$ | [24, 17, 5] | 8 | Good |
| $\langle f_2, uvf_1, v^2f_1f_3\rangle$ | [24, 21, 2] | 4 | Good |
| $\langle f_1, uvf_1f_2, (uv^2 + wv^3)f_2f_3\rangle$ | [24, 20, 3] | 5 | Good |

| Generators of cyclic codes | Images under $\theta$ | $d_{MDS}$ | Remark |
|---|---|---|---|
| $\langle vf_3, uf_1, uv^2f_2, (uv+v^2+v^3w+uv^2)f_2f_3\rangle$ | [24, 19, 4] | 6 | Good |
| $\langle uvf_2, uf_3, uv^2f_1f_2, (uv^2+v^3)f_1f_3\rangle$ | [24, 16, 6] | 9 | Good |
| $\langle f_2, vf_1, uvf_2f_3\rangle$ | [24, 22, 2] | 3 | Good |

***Example 2*** For $n = 5$, we have $x^5 - 1 = (x+1)(x+w^3)(x+w^6)(x+w^9)(x+w^{12}) \in \mathbb{F}_{16}[x]$. Let $f_1 = x+1$, $f_2 = x+w^3$, $f_3 = x+w^6$, $f_4 = x+w^9$, $f_5 = x+w^{12}$. Then some cyclic codes of length 5 over $R$ and their Gray images are given below.

| Generators of cyclic codes | Images under $\theta$ | $d_{MDS}$ | Remark |
|---|---|---|---|
| $\langle f_2, uf_3, uvf_4f_5\rangle$ | [40, 37, 2] | 4 | Good |
| $\langle uf_1, vf_2f_3, uvf_4f_5, (uv^2+v^3)f_1f_3f_5\rangle$ | [40, 32, 6] | 9 | Good |
| $\langle vf_3, uvf_5, uv^2f_1f_2\rangle$ | [40, 34, 4] | 7 | Good |
| $\langle f_2f_4, vf_3, uvf_1f_4, (uv+v^2w+v^3+uv^2w^2)f_1f_3f_5\rangle$ | [40, 35, 3] | 6 | Good |

## 7 Conclusion

In 2019, the structure of cyclic codes over $M_4(\mathbb{F}_2)$ was discussed by Pal et al. [13] for odd length. Initially, we have extended their work to the matrix ring $R = M_4(\mathbb{F}_2 + u\mathbb{F}_2)$ and derived the structure theorem for cyclic codes of odd length. Then we have obtained the images of cyclic codes of odd length over the ring $R$ under $\phi$ as binary quasi-cyclic codes of even length over $M_4(\mathbb{F}_2)$. Based on derived results, we have provided some good linear codes over $\mathbb{F}_{16}$. Here, a natural question comes to generalizing this study of codes of even length over $R$ and finding their generators. It is still open and we can carry forward this research to both the Euclidean and the Hermitian self-dual cyclic codes over the ring $R = M_4(\mathbb{F}_2 + u\mathbb{F}_2)$.

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